

In a civil partnership, happy father of three.

Research interests (2017–)

Monte Carlo methods; interacting particle systems; probabilistic modelling and inference; quantum computing; applications to signal processing, machine learning, quantum optics, and cardiac cell biology.

Academic positions

- 2024– **Directeur de recherche CNRS (DR2)**, *CRIStAL, Université de Lille*, Lille (France)
CRIStAL [link] is the dept. of computer science, signal processing and automatic control. I am also partially affiliated with *École Centrale de Lille*, a top-tier engineering school in France.
- Jan.–Jul. 2023 **Invited researcher**, *Universitat Pompeu Fabra*, Barcelona (Spain)
Visiting both the *statistics group* [link] and the *artificial intelligence and machine learning group* [link].
- Feb. 2015 – **Chargé de recherche CNRS (CRCN)**, *CRIStAL, Université de Lille*, Lille (France)
Oct. 2024 *CRIStAL* [link] is the dept. of computer science, signal processing and automatic control.
- 2013–2015 **Postdoctoral fellow**, *Department of Statistics, University of Oxford*, Oxford (UK)
2020 Science fellowship of the EPSRC, working with Chris Holmes on large-scale Markov chain Monte Carlo methods, motivated by biological data analysis.

Education

- 2022 **Habilitation à diriger des recherches (HDR)**, *Université de Lille*, Lille (France)
Computer engineering, automatic control, and signal processing, *Turning repulsive point processes into subsampling algorithms*. Reviewers were *Patrick Flandrin (ENS Lyon)* [link], *Arthur Gretton (University College London)* [link], and *Judith Rousseau (Univ. Oxford)* [link].
- 2009–2012 **Doctorat (Ph.D.)**, *Université Paris-Sud XI, Orsay* (France), *highest honours*
Computer science, *Towards adaptive learning and inference - Applications to hyperparameter tuning and astroparticle physics*, under the supervision of *Balázs Kégl* [link]. My thesis reviewers were *Éric Moulines* [link] and *Christian Robert* [link].
- 2008–2009 **Master (M.Sc.)**, *Ecole Normale Supérieure, Cachan* (France), *highest honours*
Mathematics, computer vision and machine learning ('MVA').
- 2008 **Agrégation**, *Université Louis Pasteur, Strasbourg* (France)
Mathematics major, probability & statistics minor.

Recent (2017–) management of funding resources

- 2025–2029 **WP leader of a Univ. Lille cross-disciplinary programme, 3.2 M€**
Project C2EMPI “Complexity in the physical world and Emergent Mathematical tools”. The programme consists of 100+ people across four labs (roughly speaking computer science, maths, laser physics, and electronics). I lead one of seven WPs, on *disordered systems, from probability to signal processing and material science*.
- 2020–2025 **PI of a national individual chair in artificial intelligence, 880 k€**
Project BACCARAT “Bayesian learning of expensive models, with applications to cell biology”.
- 2020–2026 **PI of an ERC Starting Grant, 1.5 M€**
Project BLACKJACK “Fast Monte Carlo integration with repulsive processes”.
- 2017–2021 **PI of an LAI, 20 k€**
Project TUNNELING EFFECT between Univ. Lille and Univ. Oxford (UK).
LAIs (“associated international labs”) are initiatives from Univ. Lille to consolidate existing strong international scientific collaborations. They are also sésames to PhD funding from the university.
- 2016–2020 **PI of an ANR JCJC grant, 172 k€**
Project BoB “Bayesian inference on a budget”.

Awards

- 2021 **CNRS bronze medal**, *CNRS Institute for information sciences and their interactions (INS2I)*
CNRS bronze medals [link] distinguish the early careers of promising researchers working in France. Each of the thematic institutes that form CNRS awards a handful (≤ 3 for INS2I) of medals per year.
- 2013 **2nd prize of the Gilles Kahn award**, *Société Informatique de France*
 Awarded by the French society of Computer Science for the best French Ph.D. of the academic year.

Recent (2017–) international seminars and visits

- ▷ Statistics seminar, Univ. Warwick, **Coventry, UK** (2025).
- My third kid was born at the end of 2023, and I have been limiting my travels since. Also, I now tend to refuse invitations that involve flying for carbon footprint reasons.
- ▷ Statistics seminar, Universitat Pompeu Fabra, **Barcelona, Spain** (2023).
- ▷ Analysis seminar, Universitat de Barcelona, **Barcelona, Spain** (2023).
- ▷ Machine learning seminar, Universitat Pompeu Fabra, **Barcelona, Spain** (2023).
- ▷ Dept. of decision sciences, Bocconi University, **Milan, Italy** (1 week, 2021).
- ▷ Machine learning group, CWI, **Amsterdam, Netherlands** (1 week, 2019)
- ▷ European space agency technical labs, **Leiden, Netherlands** (2 days, 2019)
- In 2018, my second kid was born and I limited my international travels to a minimum.
- ▷ Dept. of mathematics, **Univ. Bristol, UK** (3 days, 2017)
- ▷ Dept. of mathematics, **Univ. Aalborg, Denmark** (1 week, 2017)

Recent (2017–) national seminars and visits

- ▷ Dept. of probability and statistics (LPSM), **Sorbonne Université**, Paris (2024, one day).
- ▷ Dept. of mathematics, **Univ. Paris-Saclay**, Orsay (2024, one day).
- ▷ Center for innovation in telecommunications and integration of service (CITI), **INSA Lyon**, Lyon (2023, one day).
- ▷ Dept. of mathematics, **Univ. Paris-Descartes**, Paris (2019, one day).
- ▷ Dept. of mathematics, **Univ. Paris-Saclay**, Orsay (2018, one day).
- ▷ *Journée algorithmes stochastiques* [link], **Univ. Paris-Dauphine**, Paris (2017, one day).
- ▷ Dept. of mathematics *Jean Leray*, **Univ. Nantes**, Nantes (2017, two days).

Recent (2017–) invited talks at international workshops or international conferences

- ▷ Plenary speaker at a the conference on *Sampling theory and applications (SampTA)*, Univ. Vienna, Austria, July 2025.
- ▷ Invited speaker at a symposium on point processes, Univ. Twente, Enschede, Netherlands, February 2025.
- ▷ Invited speaker in a contributed session at the International conference on statistics and data science (ICSIDS), Nice, France, December 2024.
- ▷ Invited speaker at the *London meeting on computational statistics* [link], London, UK, June 2024.
- ▷ Invited speaker at the *Dutch meeting of probabilists and statisticians* [link] Lunteren, Netherlands, November 2022.
- ▷ Plenary speaker at the *International conference on belief functions* [link], Sorbonne University, Paris, October 2022.
- ▷ Plenary speaker at the *Workshop on kernel approximations and space-filling* [link], Cardiff University, Cardiff, July 2022.
- ▷ Invited speaker at the *Workshop on inference for expensive systems in mathematical biology* [link], University of Oxford, UK, May 2022.
- ▷ Plenary speaker at *Mathematical statistics and learning* [link], Univ. Pompeu Fabra, Barcelona, June 2021.
- postponed Plenary speaker at the international conference on *Mathematical Methods in Physics* [link], Marrakech, Morocco.
- online ICML workshop on *Negative dependence in machine learning*, International Conference on Machine Learning (ICML), July 2020 (held online).
- cancelled Session on *Kernel methods for uncertainty quantification*, SIAM conference on uncertainty quantification, Munich, March 2020.
- ▷ Workshop on *Stochastic Geometry, Stereology and Image Analysis*, Sandbjerg, Denmark, June 2019.
- ▷ Workshop on *Point processes in Space, Time and Beyond*, Skagen, Denmark, May 2019.

- ▷ *Workshop* [link] on *Bayesian nonparametrics for signal/image processing*, Bordeaux, France, July 2018.
- ▷ *MLSS African machine learning summer school*, Algiers, Algeria, June 2018.
- ▷ *SSP conference on Statistical signal processing*, Freiburg, Germany, June 2018.
- ▷ *Workshop on Bayesian nonparametrics for signal/image processing*, Bordeaux, France, June 2018.
- ▷ *Workshop on Cardiac modelling* of the Royal Statistical Society, Chicheley, UK, February 2017.

Selected recent (2017–) invited talks in national workshops/conferences

- ▷ Tutorial speaker at the annual *MASCOT-NUM meeting* [link] on uncertainty quantification, ENSAI Rennes, France, April 2026.
- ▷ Invited speaker at the workshop on Kirchhoff forests, Centre international de rencontres math matiques (CIRM), Univ. Aix-Marseille, France, November 2025.
- ▷ Plenary speaker at the annual conference of the French statistical society (*Journ es de statistiques de la SFdS*) [link], Bordeaux, France, May 2024.
- ▷ Invited speaker at the workshop *Statistiques au sommet* [link], M g ve, France, March 2024.
- ▷ Tutorial speaker at *Stochastic geometry days* [link], Univ. du Littoral C te d'Opale, Dunkerque, France, November 2021.
- ▷ *Workshop on kernel and sampling methods* [link] of GDR Mascot-Num, Institut Henri Poincar , Paris (2021, 1 day).
- ▷ Tutorial speaker at *End-to-End Bayesian learning* [link] Centre international de rencontres math matiques (CIRM), Univ. Aix-Marseille, France, October 2021.
- ▷ Invited speaker at the French *Acad mie des Sciences* for a mini-workshop on determinantal point processes, June 2018.
- ▷ Invited speaker at the Physics colloquium of ENS Lyon, France, March 2018.
- ▷ Plenary speaker at the *StatLearn* [link] workshop, Univ. Lyon, April 2017.
- ▷ Plenary speaker at the *GRETSI* [link] conference, September 2017.
- ▷ Bayes in Paris [link] *national seminar series*, Paris, April 2016.

Recent (2017–) teaching experience

CNRS positions come with no teaching duty. Out of personal inclination, I still maintain a small teaching activity at the master level.

- 2019– **Lecturer**, *ENS Paris-Saclay*, Cachan, then Saclay (France)
16 hours per year, on Bayesian machine learning (“MVA” master programme; the most demanded machine learning master nationwide.)
- 2019–2022 **Lecturer**, *Univ. Lille and  cole Centrale de Lille*, Lille (France)
18 to 24 hours per year, on Model-based machine learning (master-level students in data science).
- 2016–2019 **Lecturer**, *ENSAE ParisTech*, Paris (France)
9 hours per year, on Bayesian nonparametrics (master-level students in statistics and econometry).
- 2015–2019 **Lecturer**, * cole centrale de Lille*, Lille (France)
15 hours per year, on practical machine learning with applications to bankruptcy prediction (master-level engineering students). I organized this course around a machine learning competition, see e.g. the *2019 edition* [link].

Recent (2017–) scientific responsibilities and research management

- 2026 co-organizing *Point processes and applications* [link], a three-day international conference in Lille, France.
- 2024– Head of team SigMA (*Signals, models, and applications* [link]) within my lab CRIS AL. SigMA corresponds to ~35 people out of which 15 permanent members, across 4 employers.
- 2023 co-organized *Bayes at CIRM'23* [link], a one-week autumn school on *Bayesian statistics* [link] at CIRM Marseille, France.
- 2022– Associate Editor for *Bayesian Analysis* [link].
- 2022 co-organized a one-week international workshop on *Computational methods for unifying multiple statistical analyses* [link] at CIRM Marseille, France.
- 2022 co-organized the *second workshop on DPPs and fermions* [link], at ENS Lyon, France. [This two-week workshop](#), my longest organized event so far, was a follow-up to our international *first workshop on DPPs and fermions* [link] in Lille, France. across physics, mathematics, and signal processing.

- 2019–2020 Jury member for the French PhD prize in *signal & image processing and computer vision* [link]. Served as president in 2020 (mandates are one-year).
- 2019 co-organised *GRETSI* [link] at Univ. Lille, the main French-speaking event in signal processing. [This is my biggest event so far \(~450 participants\)](#).
- 2019 co-organised the first international *workshop on DPPs and fermions* [link] across physics, mathematics, and signal processing at Univ. Lille.
- 2017–2018 I ran a small workgroup on *reproducible research* in Lille, where software engineers transfer selected skills to researchers to achieve high standards of reproducibility. See our *tutorial* [link] for instance, or a *concrete example* [link] of our standards: our software package on DPP sampling for machine learning. More software activity on my *GitHub account* [link]
- 2017– Reviewer for journals such as *Annals of Statistics*, *Annals of Applied Probability*, *Journal of Machine Learning Research (JMLR)*, *Journal of the Royal Statistical Society B*, *Biometrika*, *Bernoulli*, *Journal of Econometrics*, *IEEE Transactions on Signal processing (TSP)*, *Bayesian Analysis*, *Statistical Science*, *Electronic Journal of Statistics (EJS)*, *Journal of Computational and Graphical Statistics (JCGS)*, *Statistics and Computing*,
- 2017– Reviewer and occasional area chair for conferences such as *International Conference on Machine Learning (ICML)*, *Advances in Neural Information Processing Systems (NeurIPS)*, *International conference on learning theory (COLT)*..

Supervision

Percentages display my own share of the supervision. I am proud that all my postdocs obtained an academic or industrial research position (in maths, information sciences, or physics).

- 2025– **Postdoc**, CNRS, Lille (France), 50%
I am co-supervising Quentin François's Ph.D. with Raphaël Butez (MCF Univ. Lille & Laboratoire Painlevé), on *Identifying the support of the signal in the point process of zeros of its spectrogram*.
- 2024– **Ph. D. student**, Ecole Centrale de Lille, Lille (France), 50%
I am co-supervising Vladimir Petrovic's Ph.D. with Agnès Desolneux (DR CNRS at Centre Borelli, ENS Paris-Saclay), on *Determinantal point processes and optimal transport*.
- 2023–2024 **Postdoc**, CNRS, Lille (France), 100%
I supervised Juan Manuel Miramont's postdoc on *Topological data analysis applied to time-frequency analysis*. Juan is now an assistant professor in engineering at Institut Mines-Télécom Nord-Europe, Lille, France.
- 2023–2024 **Postdoc**, CNRS, Lille (France), 100%
I supervised Thibaut Lemoine's postdoc on *Monte Carlo integration on manifolds*. Thibaut then obtained a fellowship at Collège de France, Paris, France. He is now an assistant professor in mathematics at Univ. Strasbourg, France.
- 2022–2026 **Ph. D. student**, CNRS, Lille (France), 50%
I am co-supervising Martin Rouault's Ph.D. with Mylène Maida (Prof. Univ. Lille & Laboratoire Painlevé), on *Monte Carlo integration with Riesz gases*.
- 2021–2023 **Postdoc**, Univ. Nantes, Nantes (France), 30%
I have co-supervised Juan Manuel Miramont's postdoc with François Auger (Univ. Nantes), Pierre Chainais (Univ. Lille), and Patrick Flandrin (ENS Lyon), on *Zeros of the Fourier transform of random signals*. Juan Manuel then continued on a different postdoc with me at CRISAL.
- 2021–2022 **Postdoc**, CNRS, Lille (France), 100%
I supervised Xiaoyi Mai's postdoc on *Repulsive point processes in machine learning*. Xiaoyi then obtained a assistant professorship in mathematics at Univ. Toulouse.
- 2021–2025 **Postdoc**, CNRS, Lille (France), 100%
I supervised Michaël Fanuel's postdoc on *Repulsive point processes for inverse problems*. Michaël is now a research engineer at IMEC, Leuven, Belgium.
- 2021 **Research software engineer**, CNRS, Lille (France), 100%
I supervised Guillaume Gautier's work as a software research engineer in my group. Guillaume then became a data scientist at Decathlon, Lille.
- 2021–2023 **Postdoc**, CNRS, Lille (France), 100%
I have supervised Alexandre Feller's postdoc on *Determinantal point processes and fermions in quantum optics*. Alexandre then obtained an assistant professorship in physics at Univ. Lille.
- 2020–2022 **Postdoc**, CNRS, Lille (France), 100%
I supervised Barbara Pascal's postdoc on *Repulsive point processes in time-frequency analysis*. Barbara then obtained a CNRS junior permanent position at Univ. Nantes, France.

- 2020–disc. **Ph. D. student**, *Ecole Centrale de Lille*, Lille (France), 50%
I co-supervised the beginning of Yoann Mayer's Ph.D. with Mylène Maida (Prof. Univ. Lille & Laboratoire Painlevé), on *Sampling algorithms for repulsive stochastic processes*. Yoann unfortunately quit before completing his PhD to join the finance industry.
- 2020–2023 **Ph. D. student**, *Univ. Lille*, Lille (France), 50%
I have co-supervised Diala Hawat's Ph.D. with Raphaël Lachièze-Rey (HDR, Univ. Paris & MAP5), on *Repulsive points patterns in statistical signal processing*. Diala then went on a postdoc at Sorbonne Université, and is now a research engineer at EDF labs, Saclay, France.
- 2019–2021 **Postdoc**, *CNRS*, Lille (France), 100%
I supervised Arnaud Poinas's postdoc on *Repulsive point processes in experimental design*. Arnaud was co-supervised by Adrien Hardy (Univ. Lille & Laboratoire Painlevé) during his first year of postdoc. Arnaud then obtained an assistant professorship in mathematics at Univ. Poitiers, France.
- 2017–2020 **Ph. D. student**, *Ecole Centrale de Lille*, Lille (France), 50%
I co-supervised Ayoub Belhadji's Ph.D. with Pierre Chainais (Prof. Centrale Lille & CRIStAL), on *Determinantal point processes for dimension reduction in signal processing*. Ayoub went on to a postdoc with Rémi Gribonval at ENS Lyon, France, and then to another postdoc with Youssef Marzouki at MIT, Cambridge, USA.
- 2016–2020 **Ph. D. student**, *Ecole Centrale de Lille*, Lille (France), 50%
I co-supervised Guillaume Gautier's Ph.D. with Michal Valko (HDR, Inria Lille & CRIStAL), on *Fast sampling of determinantal point processes*. Guillaume went on to a postdoc with Pierre-Olivier Amblard, Simon Barthelmé, and Nicolas Tremblay at Univ. Grenoble-Alpes, France.
- 2014–2018 **Ph. D. student**, *Univ. Oxford*, Oxford (UK), 20%
I have co-supervised Ross Johnstone's Ph.D. with Gary Mirams and David Gavaghan (Oxford Computer Science), with industrial collaborators at *Roche labs* (Basel, Switzerland) on *Uncertainty characterisation in action potential modelling for cardiac drug safety*. Ross went on to an R&D position in data science at Rakuten, Japan.

Recent (2017–) track Record

In this section, I present my research interests in a loose chronological order, restricting to themes in which I have been active in the last five years.

Scalable Bayesian inference. Loosely speaking, Bayesian inference for big data faces two main challenges: big data can be *tall* – it contains an overwhelming number of individual data points – or it can be *large* – each data point contains a large number of components. Since 2013, I have had a keen interest into scaling up Bayesian inference to tall datasets while maintaining the theoretical properties that make Bayesian inference desirable in the first place. Restricting to the period 2017–2021, I have proposed the first MCMC algorithm that can provably use less than the original number of data points while maintaining the same accuracy as if using the entire dataset [17]. Focusing on large datasets, this time, we obtained efficient reduction of dimensionality in linear regression using randomized column subset selection [11]. For a while now, I have been thinking of tall data and costly Bayesian inference from a more foundational point of view, trying to understand what coherent decision making means in this context; more to appear soon on this topic.

Applications to biology. I have been part of the 2020 science network – a UK-based network of young computational scientists. In the 2017–2021 time frame, I have transferred statistical tools to computational biology, in particular cardiac modelling [53]. After that, I have been collaborating with the US FDA and its British equivalent to assess the dangerous side-effects of anti-arrhythmia drugs [52].

Determinantal point processes. This has been my most active recent theme. Determinantal point processes (DPPs) are point process models that naturally encode diversity or repulsiveness between the points of a given realization, using algebraic arguments. They arise in many fields, e.g. random matrix theory, combinatorics, or quantum physics. Our landmark result was to show that DPPs lead to fast Monte Carlo integration [12], leading to a natural stochastic version of Gaussian quadrature, and we have proven faster-than-Monte-Carlo central limit theorems for our estimators. See also [39] for a different description of the train of thought that took us to the results in [12]. Another line of attack of the same problem led us to generic bounds for numerical integration and signal sampling in reproducing kernel Hilbert spaces using DPPs [25, 24, 1]. In a similar spirit, we are now investigating DPPs for experimental design [8], and subsampling in stochastic gradient methods [23]. Finally, all these applications of DPPs require efficient sampling algorithms, to which we have contributed both in methodology [28, 26] and software [14]. Still on DPP sampling, we have recently been interested in how to do it on a quantum computer [6]; this turns out to be a natural task for a quantum computer.

Applications to signal processing. After a stimulating *talk* [link] by Patrick Flandrin, I was intrigued by the seemingly repulsive point process formed by the zeros of the short-time Fourier transform of white noise. We proved [13] that this process is an instance of zeros of a Gaussian analytic function, an object of recent booming interest in the random matrix community. This opened a probabilistic toolbox full of theorems that apply directly to the design of filters for signal processing. This first result sparked a lot of interest and raised new deep questions. In our follow-up work [15], we gave new links to an analytic wavelet transform as well as discrete windowed Fourier transforms. This gives yet another viewpoint and a stimulating new field of application for repulsive point processes such as DPPs. These questions are part of the new ANR project ASCETES (PI Sylvain Meignen), to which I'm participating. Currently, I am also revisiting the links of the formalisms of time-frequency analysis and quantum optics, putting the involved point processes at the forefront [63]. In a sense, this is a modern re-reading of Odile Macchi's rich 1975 thesis, where she defined DPPs to model the detection times of a beam of free fermions.

Publications

My research is multidisciplinary, and publishing habits differ among the relevant disciplines, like in probability, statistics, signal processing, cell biology, and experimental physics. I group computer science, computational statistics, applied probability, and signal processing under the acronym 'CS' below. You can also check my Google Scholar *profile* [link].

CS journals

- [1] A. Belhadji, **R. Bardenet**, and P. Chainais. Signal reconstruction using determinantal sampling. *Applied and Computational Harmonic Analysis (ACHA)*, 82, 2026.
- [2] M. Fanuel and **R. Bardenet**. Cycling in the forest with Wilson's algorithm. *Probability Surveys*, 2026.
- [3] V. Petrovic, **R. Bardenet**, and A. Desolneux. Repulsive Monte Carlo on the sphere for the sliced Wasserstein distance. *Transactions on Machine Learning Research (TMLR)*, 2026.
- [4] J. M. Miramont, **R. Bardenet**, P. Chainais, and F. Auger. mcsbm-benchs: Benchmarking methods for multi-component signal processing. *Journal of Open Source Software*, 2025.
- [5] M. Fanuel and **R. Bardenet**. Sparsification of the regularized magnetic Laplacian with multi-type spanning forests. *Applied and Computational Harmonic Analysis (ACHA)*, 78, 2025.
- [6] **R. Bardenet**, M. Fanuel, and A. Feller. On sampling determinantal and Pfaffian point processes on a quantum computer. *Journal of Physics A: Mathematical and Theoretical*, 2024.
- [7] D. Hawat, G. Gautier, **R. Bardenet**, and R. Lachièze-Rey. On estimating the structure factor of a point process, with applications to hyperuniformity. *Statistics and Computing*, 33, 2023.
- [8] A. Poinas and **R. Bardenet**. On proportional volume sampling for experimental design in general spaces. *Statistics and Computing*, 2022.
- [9] B. Pascal and **R. Bardenet**. A covariant, discrete time-frequency representation tailored for zero-based signal detection. *IEEE Transactions on Signal Processing*, 2022.
- [10] G. Gautier, **R. Bardenet**, and M. Valko. Fast sampling of β -ensembles. *Statistics and Computing*, 2021.
- [11] A. Belhadji, **R. Bardenet**, and P. Chainais. A determinantal point process for column subset selection. *Journal of Machine Learning Research (JMLR)*, 2020.
- [12] **R. Bardenet** and A. Hardy. Monte Carlo with determinantal point processes. *Annals of Applied Probability*, 2020.
- [13] **R. Bardenet**, J. Flamant, and P. Chainais. On the zeros of the spectrogram of white noise. *Applied and Computational Harmonic Analysis*, 2020.
- [14] G. Gautier, **R. Bardenet**, G. Polito, and M. Valko. DPPy: Sampling determinantal point processes with Python. *Journal of Machine Learning Research; Open Source Software (JMLR MLOSS)*, 2019.
- [15] **R. Bardenet** and A. Hardy. Time-frequency transforms of white noises and Gaussian analytic functions. *Applied and Computational Harmonic Analysis*, 2019.
- [16] **R. Bardenet**, F. Lavancier, X. Mary, and A. Vasseur. On a few statistical applications of determinantal point processes. *ESAIM: Proceedings and Surveys*, 60:180–202, 2017.
- [17] **R. Bardenet**, A. Doucet, and C. Holmes. On Markov chain Monte Carlo methods for tall data. *Journal of Machine Learning Research (JMLR)*, 2017.
- [18] **R. Bardenet** and O.-A. Maillard. Concentration inequalities for sampling without replacement. *Bernoulli*, 2015.
- [19] **R. Bardenet**, O. Cappé, G. Fort, and B. Kégl. Adaptive MCMC with online relabeling. *Bernoulli*, 21(3):1304–1340, 2015.

CS refereed international proceedings

- [20] **R. Bardenet**, S. Ghosh, H. Simon-Onfroy, and H.-S. Tran. Small coresets via negative dependence: DPPs, linear statistics, and concentration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [21] H. Jaquard, M. Fanuel, P.-O. Amblard, **R. Bardenet**, S. Barthelmé, and N. Tremblay. Smoothing complex-valued signals on graphs with Monte-Carlo. In *Proceedings of the International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023.
- [22] M. Fanuel and **R. Bardenet**. Nonparametric inference of continuous DPPs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [23] **R. Bardenet**, S. Ghosh, and M. Lin. Determinantal point processes based on orthogonal polynomials for sampling minibatches in SGD. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [24] A. Belhadji, **R. Bardenet**, and P. Chainais. Kernel interpolation with continuous volume sampling. In *International Conference on Machine Learning (ICML)*, 2020.
- [25] A. Belhadji, **R. Bardenet**, and P. Chainais. Kernel quadrature with determinantal point processes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [26] G. Gautier, **R. Bardenet**, and M. Valko. On two ways to use determinantal point processes for Monte Carlo integration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [27] **R. Bardenet**, P. Chainais, J. Flamant, and A. Hardy. A correspondence between zeros of time-frequency transforms and Gaussian analytic functions. In *Proceedings of the International conference on Sampling Theory and its Applications (SampTA)*, 2019.
- [28] G. Gautier, **R. Bardenet**, and M. Valko. Zonotope hit-and-run for efficient sampling of projection dpps. In *International Conference on Machine Learning (ICML)*, 2017.
- [29] **R. Bardenet** and M. K. Titsias. Inference for determinantal point processes without spectral knowledge. In *Advances in Neural Information Processing Systems (NIPS)*, pages 3375–3383, 2015.
- [30] **R. Bardenet**, A. Doucet, and C. Holmes. Towards scaling up MCMC: an adaptive subsampling approach. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2014.
- [31] **R. Bardenet**, M. Brendel, B. Kégl, and M. Sebag. Collaborative hyperparameter tuning. In *International Conference on Machine Learning (ICML)*, 2013.
- [32] **R. Bardenet**, O. Cappé, G. Fort, and B. Kégl. An adaptive Metropolis algorithm with online relabeling. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 22, pages 91–99, April 2012.
- [33] J. Bergstra, **R. Bardenet**, B. Kégl, and Y. Bengio. Algorithms for hyperparameter optimization. In *Advances in Neural Information Processing Systems (NIPS)*, volume 24. The MIT Press, 2011.
- [34] **R. Bardenet** and B. Kégl. Surrogating the surrogate: accelerating Gaussian-process-based global optimization with a mixture cross-entropy algorithm. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 2010.

CS refereed national proceedings (selected papers)

- [35] X. Mai and **R. Bardenet**. Un processus ponctuel déterminantal pour la sélection de variables supervisée. In *Actes de la conférence du GRETSI*, 2022.
- [36] B. Pascal and **R. Bardenet**. Une famille de représentations covariantes de signaux discrets et son application à la détection de signaux à partir de leurs zéros. In *Actes de la conférence du GRETSI*, 2022.
- [37] J. M. Miramont, **R. Bardenet**, P. Chainais, and F. Auger. A public benchmark for denoising and detection methods. In *Actes de la conférence du GRETSI*, 2022.

CS refereed workshops

- [38] G. Gautier, **R. Bardenet**, and M. Valko. On two ways to use determinantal point processes for Monte Carlo integration. Technical report, ICML workshop on Negative dependence in machine learning, 2019.
- [39] **R. Bardenet** and A. Hardy. From random matrices to Monte Carlo integration via Gaussian quadrature. In *Proceedings of the IEEE Statistical Signal Processing workshop (SSP)*, 2018.

CS book chapters

- [40] **R. Bardenet**. *Proceedings of the 2012 IN2P3 School of Statistics*, chapter Monte Carlo methods. EDP Sciences, 2013.

Physics Journals (selected papers)

- [41] P. Abreu et al. The lateral trigger probability function for UHE cosmic rays showers detected by the Pierre Auger Observatory. *Astroparticle Physics*, 35:266–276, 2011. Auger Collaboration paper.
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